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Research Paper

Vibration-induced heat transfer enhancement in additively manufactured Kelvin metal foam

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- A new full scale Kelvin foam compact heat exchangers was additively fabricated using AlSi10Mg powder.
- The performance of metal foam heat exchangers at 1mm~5mm and 20Hz~40Hz was tested and simulated.
- The comprehensive performance improvement of active and passive enhanced heat transfer is analyzed.
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Abstract

Compact porous media heat exchanger presents a promising solution for advanced thermal management in vehicles. However, their heat transfer performance under inevitably vibration conditions is not yet well understood. In this regard, this paper studies the heat transfer enhancement mechanisms of Kelvin metal foam heat exchanger induced by vibration via experiments and pore scale simulations. An $80\,mm \times 270\,mm \times 210\,mm$ full scale Kelvin metal foam heat exchanger was fabricated using AlSi10Mg powder through selected laser melting (SLM) additive manufacturing. The effect of vibration amplitude and frequency on its performance was experimentally studied using a wind tunnel with a vibration platform. Results indicate that (i) vibration can improve the performance of the heat exchanger, with a maximum effective heat transfer coefficient (*EHTC*) of $1.56\,kW/(m^2{\cdot}K)$, a 13.93% increase over the $1.37\,kW/(m^2{\cdot}K)$ observed without vibration; (ii) The Analysis of Variance (ANOVA) reveals that amplitude and frequency similarly affect its heat transfer performance within the vibration range of 1–5 mm and 20–40 Hz; (iii) Vibration enhances heat transfer by breaking the flow boundary layer. The heat transfer coefficient varies periodically with vibration and increases as vibration amplitude rises. Notably, the change frequency of heat transfer coefficient is twice that of the vibration frequency. This work quantitatively analyzes how

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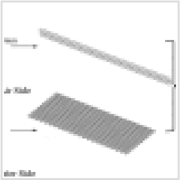
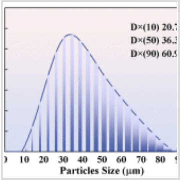
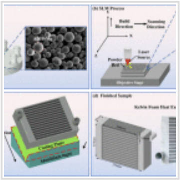
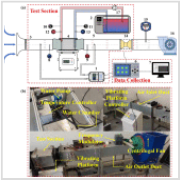
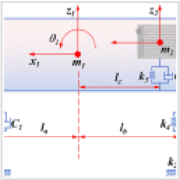
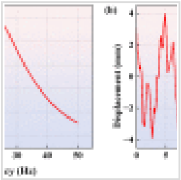

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vibration improves the performance of metal foam heat exchanger and offers significant data reference for their practical applications.

Keywords

Kelvin metal foam heat exchanger; Vibration conditions; Laser additive manufacturing; Conjugate heat transfer; Force convection

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Nomenclature

A
: Area (m^2)

c
: Damping coefficient (N·s/m)

C_p
: Specific heat capacity (J/kg·K)

d
: Diameter (mm)

D
: Diameter (mm)

f
: Frequency (Hz)

F

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Friction factor

H

Height of air side

h

Heat transfer coefficient

J

Colburn factor

k

Stiffness (N/m)

l

Distance (m)

L

Distance (m)

$\dot{m}$

Mass flow rate (kg/s)

Nu

The Nusselt Number

Pr

The Plante Number

P

Pressure (Pa)

q

Road excitation (m)

Q
: Heat quantity (J)

Re
: The Reynold number

t
: Time (s)

T
: Temperature (K)

v
: Velocity (m/s)

V
: Volumetric flow rate (m^3/s)

Greek Symbols

〈α_{sf}
: Specific surface area (m^2/m^3)

$\triangle$
: Differences of variable

$\triangle T_{ml}$
: The logarithmic mean temperature difference

η
: Comprehensive heat transfer factor

λ
: Thermal conductivity (W/(m·K))

υ
: Dynamic viscosity (m^2/s)

ρ
: Density (kg/m^3)

Subscrips

air
: Air

base
: Thermal Conductive Board

f
: Fluid

flow
: Heat exchanger channel for air flow

in
: Inlet

out
: Outlet

sim
: Simulation

total
: Summation of *a* and *b*

w
: Liquid working medium

Abbreviations

SLM
: Selected laser melting

EHTC
: Equivalent heat transfer coefficient ($W/m^2·K$)

CHT
: Conjugate heat transfer

1. Introduction

Metal foams, characterized by their remarkable thermal, mechanical, and acoustic properties [[1], [2], [3]], have attracted considerable attention for their potential to improve heat transfer in a range of applications, including aerospace engineering [4], lightweight structural design [5], sound insulation [6], fire prevention [7], and advanced cooling systems [[8], [9], [10]]. Metal foams improve heat transfer due to their high specific surface area, complex internal architecture, and constant disruption of thermal and viscous boundary layers [11]. The characteristics make metal foams promising candidates for next-generation vehicle heat exchanger. The use of metal foam is considered as a passive enhancement technology for vehicle heat exchanger [12]. Meanwhile, the engine intrinsically vibrates during vehicle operation and the vibration excitation is considered an active heat transfer enhancement technology. Currently, the combined effects of these mechanisms on heat exchanger performance remain unclear. Besides, research on vibration-induced heat transfer enhancement in metal foams is scarce. Consequently, it is imperative to investigate how vibration affects heat transfer in metal foam structures.

The earliest metal foam is aluminum-based and is primarily used in aerospace and military applications. As manufacturing techniques advanced, metal foams expanded to include copper, nickel, stainless steel, and titanium alloys. Currently, metal foams have achieved broader industrial applications and showed unparalleled advantages, such as, the melting and curing process of phase change materials [13,14], boiling and condensing heat transfer [15,16], and convective heat transfer. Hetsroni et al. [17] found that 20PPI metal foam enhances natural convective heat transfer 18–20 times more than a flat plate. Feng et al. [18] showed that metal foam radiators significantly boost heat transfer under natural convection. Kim et al. [19] demonstrated that high-porosity aluminum foam improves airflow and reduces thermal resistance compared to flat-fin heat exchanger. Tjoen et al. [20] reported that at high air flow rates, aluminum foam-covered tubes have double the performance evaluation coefficient of helical fin tubes.

The vibration-induced heat transfer enhancement has also been investigated across diverse contexts. Li et al. [21] studied the enhancement of air-side heat transfer in fin-tube radiators during typical vehicle vibrations. The results indicated that vibrations could improve air-side heat transfer by 3–17%. Fu et al. [22] assessed the impact of vibrations in simplified rectangular channels under conditions analogous to automobile operation, finding that vibrations boost water heat transfer by increasing turbulence. Kim et al. [23] studied forced vibrations on fin-tube heat exchanger, showing that heat transfer improves with higher vibrational frequency, amplitude, and fin pitch, but decreases with higher air-side Reynolds numbers. Cheng et al. [24] studied how transverse oscillation affects heat transfer in cylinders and found that resonance between vortex shedding and cylinder vibrations significantly boosts the heat transfer coefficient, especially at low Reynolds numbers.

Additionally, Liu et al. [25] investigated the effects of mechanical vibrations on tubular laminar flow, demonstrating that such vibrations substantially enhanced the heat transfer coefficient, especially in low Reynolds number regions. Park et al. [26] performed experiments on oscillating circular cylinders in cross-flow, noting a significant improvement in convective heat transfer as a result of vibration-induced disturbances in boundary layer. Gau et al. [27] studied forced vibrations' impact on heat transfer in cylinders, finding that the “Lock-on effect” significantly boosted the heat transfer coefficient when oscillation frequency matched the vortex shedding frequency.

Bronfenbrener et al. [28] demonstrated that vibrations could enhance turbulent mixing, thereby improving heat transfer rates in single-phase flows. Lee and Chang [29] found that vibrations increased critical heat flux in vertical tubes by enhancing turbulent mixing near the heated surface. Karanth et al. [30] used finite difference methods to show that oscillating cylinders consistently improved heat transfer coefficients under different conditions.

Faircloth and Schaetzle [31] observed that vibrations significantly boost heat transfer from horizontal cylinders by increasing turbulence. Similarly, Sreenivasan and Ramachandran [32] found that vibrations enhance heat transfer by raising air stream turbulence. Klaczak's [33] experiments showed that forced vibrations greatly improve heat transfer efficiency in heat exchanger. Leung et al. [34] also noted that vibrations significantly increase the heat transfer coefficient in low Reynolds number flows. On the basis, Choi et al. [35] examined the air-side heat transfer coefficients of discrete plate finned-tube heat exchanger with large fin pitch, identifying substantial improvements attributable to vibrational effects. Chen and Ren [36] analyzed how fin spacing affects heat transfer and pressure drop in two-row plate fin and tube heat exchanger. They found that with optimal fin spacing and vibration conditions could improve heat transfer. While vibration shows promise for enhancing heat transfer, its use with metal foam remains insufficiently investigated.

Besides, the considerable thermal contact resistance inherent in heat exchanger fabricated from conventional metal foams leads to notable measurement inaccuracies [37]. The inflexible structure of metal foams also constrains design adaptability. Advancements in additive manufacturing have led researchers to develop various regular metal foam models, including the Suleiman model, Gibson-Ashby model, Kelvin model, and Weaire-Phelan model [[38], [39], [40]]. Notably, the Kelvin model, first proposed by Kelvin in 1887, comprises a geometric configuration of four squares and ten pentagons. The tetrahedral configuration of the Kelvin model minimizes surface area per unit volume, in accordance with the principle of surface area minimization observed in foams at equilibrium [41]. The Kelvin model is often used in simulations as an alternative to conventional metal foams due to its structural similarity and strong alignment with both experimental and simulation results [42,43]. These attributes have made the Kelvin model as the most prevalently utilized structure in this domain.

This study aims to enhance heat transfer efficiency by examining how vibration affects the thermal performance of full-scale Kelvin metal foam heat exchanger fabricated via advanced additive manufacturing techniques. This work systematically analyzes how changes in vibration frequency and amplitude affect the heat transfer properties of metal foam heat exchanger. A correlation between vibration frequency and amplitude with improved heat transfer in metal form heat exchanger is established, which provides a basis understanding of vibration-induced perturbations in metal foam. The work offers practical insights for designing and optimizing the next-generation heat exchanger.

2. Experimental system and numerical simulation

A full-scale Kelvin metal foam heat exchanger was created using SLM. A wind tunnel with a vibration platform was built to assess its flow and heat transfer characteristics. Additionally, conjugate heat transfer (CHT) simulations were conducted to analyze the heat exchanger's physical field distribution under vibration to understand how vibration affects heat transfer performance.

2.1. Metal foam heat exchanger

2.1.1. Geometry of the Kelvin foam heat exchanger

This work uses the open-cell Kelvin model as an alternative to conventional metal foam, functioning as the air-side fin within the metal form heat exchanger. To meet automotive load needs and reduce stress, a circular transition was added at each skeletal junction, creating a cylindrical frame with a 0.76 mm diameter, as illustrated in Fig. 1(a). The Kelvin cell exhibits a porosity of 0.85, a pore diameter of 5.012 mm, and a specific surface area of 700.89 m^2/m^3. Staggered fins were added on the water side, where high-temperature fluid flows, to improve heat transfer efficiency, as illustrated in Fig. 1(b). The final geometry of the Kelvin metal foam heat exchanger is presented in Fig. 1(c). The macroscopic dimensions and specific details of the heat exchanger are provided in Table 1.

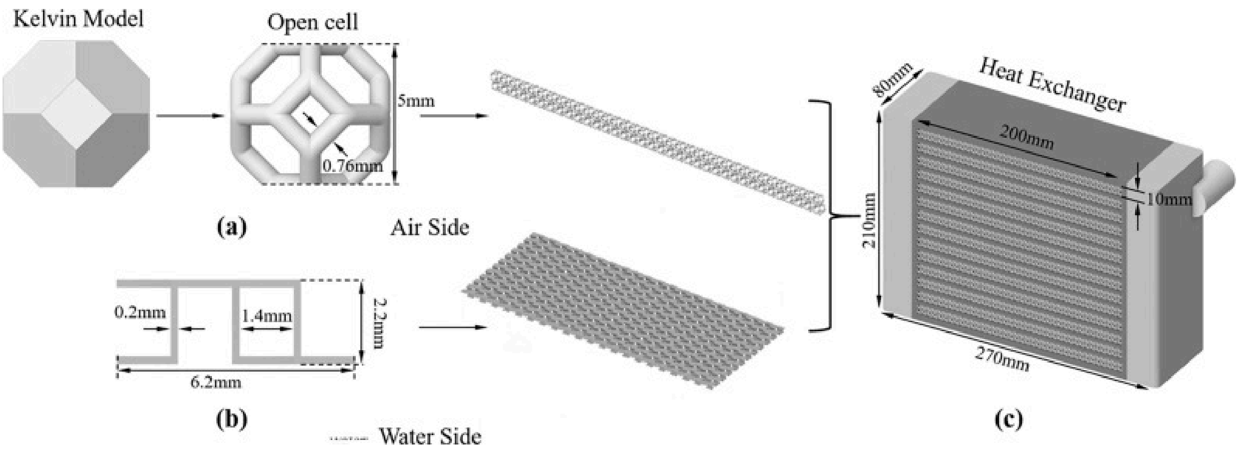

Fig. 1. Kelvin metal foam heat exchanger (a) Kelvin foam; (b) staggered fins; (c) full scale geometry.

Table 1. Geometric specifications of the Kelvin metal foam heat exchanger.

Parameters	Specifications	Parameters	Specifications
Width (mm)	270	Height of Kelvin cell (mm)	5
Length (mm)	80	Kelvin contact area (m^2)	0.095
Height (mm)	210	Kelvin foam total area (m^2)	1.944
Inner diameter (mm)	26	Air side base area (m^2)	0.448
Outer diameter (mm)	26	Air side convection area (m^2)	2.297

Parameters	Specifications	Parameters	Specifications
Fin of air side's type	Open Kelvin cell	Pore Diameter d_p	5.012
Number of air side	14	Porosity ε	0.85
Height of air side (mm)	10	Material	AlSi10Mg

2.1.2. Selective laser melting manufacturing

SLM has been widely used in creating structural components due to its precision, speed, and advanced operation. Its rapid solidification results in finer microstructures and prevents local segregation [44]. This work uses SLM with laser powder bed fusion (LPBF) to produce open-cell Kelvin metal foam heat exchanger from AlSi10Mg powder. The chemical composition of AlSi10Mg powder, expressed in weight percentage (wt%), consists of Si 9.8%, Mg 0.36%, Ti 0.042%, Cu<0.005%, Ni<0.02%, Fe 0.12%, Mn<0.005%, Zn<0.005%, Sn<0.02%, Pb<0.005%, O 0.035%, with the rest of aluminum. Fig. 2 illustrates the particle size distribution of AlSi10Mg powder. The distribution is unimodal, with an average particle size of 11.3 μm.

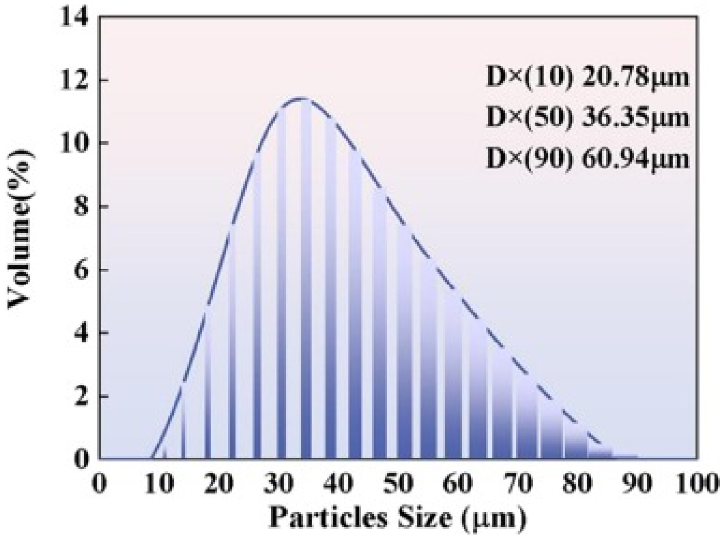

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Fig. 2. Particle size distribution of AlSi10Mg aluminum alloy powders used in this study.

The Kelvin metal foam heat exchanger was fabricated using laser selective additive manufacturing by Qingdao Sagacious Aviation Technology Co., Ltd. (China) utilizing the FS350M system (SLM Solutions, Farsson Technology Co., Ltd., China). Fig. 3 shows the additive manufacturing procedures.

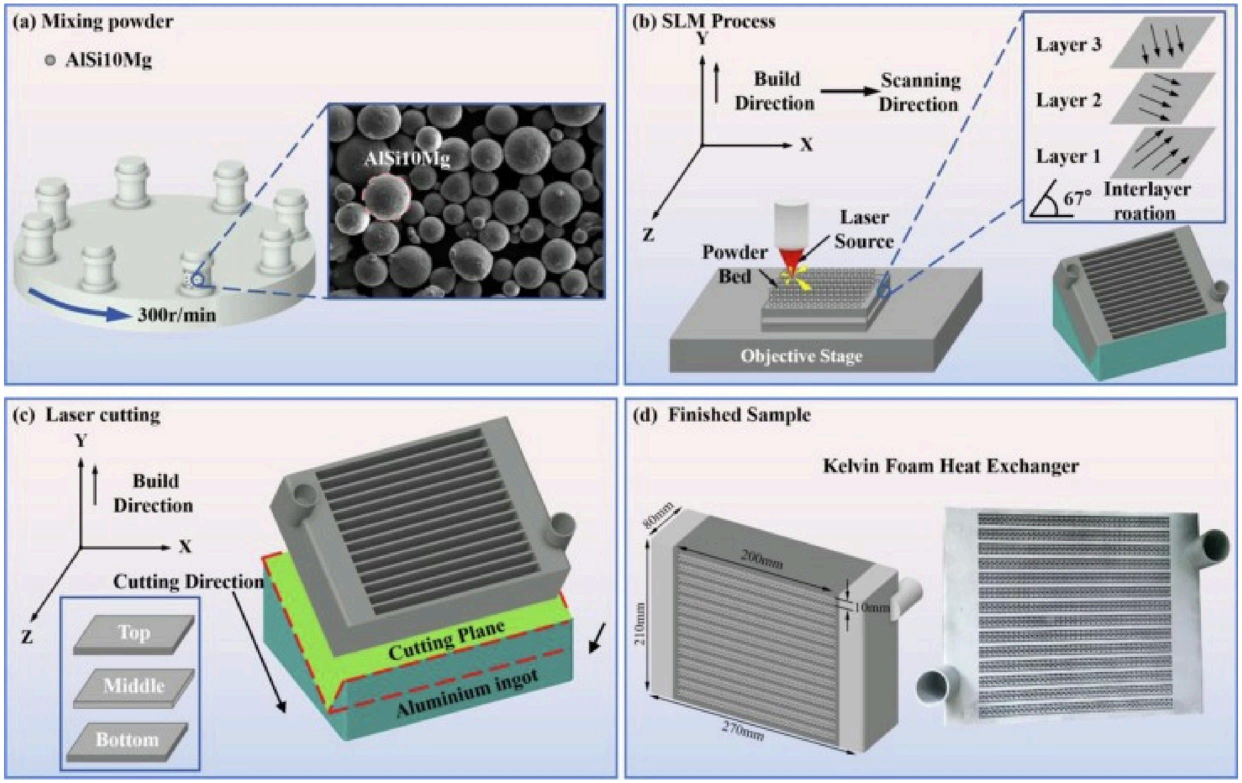

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Fig. 3. Details of SLM additive manufacturing experiment: (a) mixing powders; (b) schematic diagram and process of the SLM step; (c)wire-cutting; (d) testing samples.

Initially, the AlSi10Mg powder underwent a drying process for 6h at 328.15K. Subsequently, the residual powder, which passed through a 235-mesh screen, was combined with the new powder in a mixing machine, as illustrated in Fig. 3(a). During fabrication with AlSi10Mg powder, a continuous Yb fiber laser (λ=1080nm) with a power of 380W and a beam diameter of approximately 80μm was used. The laser scanning speed was 1.5m/s, with a scan spacing of 140μm and a layer thickness of 40μm. The pre-melt and post-melt layer thicknesses were 45μm and 35μm, respectively. The build rate was 22 mm^3/s. The structural components, following additive manufacturing conducted under argon gas with a maximum oxygen content of approximately 0.1 %, are depicted in Fig. 3(b). The scanning process used a guided scanning method with alternating angles of 67° between inter-layers. To reduce residual stress, the samples were subjected to heat treatment at 563.15 K for a duration of 2 h. Fig. 3(c) shows that the heat exchanger was wire-cut from the aluminum ingot. During demolding, a 0.2 mm grind was applied to the support surface for dimensional accuracy. The completed heat exchanger, shown in Fig. 3(d), underwent testing for vibration-enhanced heat transfer performance. The tensile and thermal conductivity properties of the AlSi10Mg aluminum alloy produced using the above process were measured using dog-bone samples, with the results shown in Table 2.

Table 2. Thermal and mechanical properties of AlSi10Mg Al alloy manufactured by SLM.

Material	**Tensile Strength [MPa]**	**Yield Strength [MPa]**	**Elongation [%]**	**Conductivity [W/(m·K)]**
AlSi10Mg	≥320	≥200	≥12	140

2.2. Force convection experimental system

Fig. 4(a) presents a schematic diagram of the experimental setup. The experimental system comprises three components-water circulation, air circulation, and vibration condition heat transfer test systems. The water circulation system comprises a constant temperature water tank, inlet and outlet pipes, a pump, a flow meter, and heat exchanger channels. The water tank acts as the heat source and has a heating capacity greater than the heat exchanger's dissipation capacity. A pump, flow control valve, and flow meter are installed at the water tank's outlet, while the pipes and tank are insulated to minimize heat loss. The air circulation system cools via an air duct, heat exchanger channels, exhaust fan, and sensors. A 210×310 mm rectangular air duct aligns with the metal form heat exchanger. Insulation cotton fills gaps to prevent the heat loss. The exhaust fan adjusts its power to provide cold air at different speeds. An airspeed device is centrally placed upstream in the air duct, with rectifier plates at both ends to ensure uniform airflow, following JB/T 10379–2022 standards [45]. Pressure is measured 520 mm from the heat exchanger, equivalent to 2 pipe diameters downstream of the test section. To precisely measure the air-side temperature at the heat exchanger, ten calibrated thermal resistances are evenly distributed at the heat exchanger inlet and outlet according to GB/T 27698. The vibration test stand is outfitted with an electromagnetic shaker that provides six degrees of freedom, functioning within a frequency range of 1 to 600 Hz, an amplitude range of 0 to 5 mm, and a maximum acceleration of 20 g. The metal foam heat exchanger is firmly affixed to this vibrating platform. The test section and the air duct are flexibly connected to avoid vibration transmission to the air duct and damage to the experimental device. Table 3 presents the main experimental equipment.

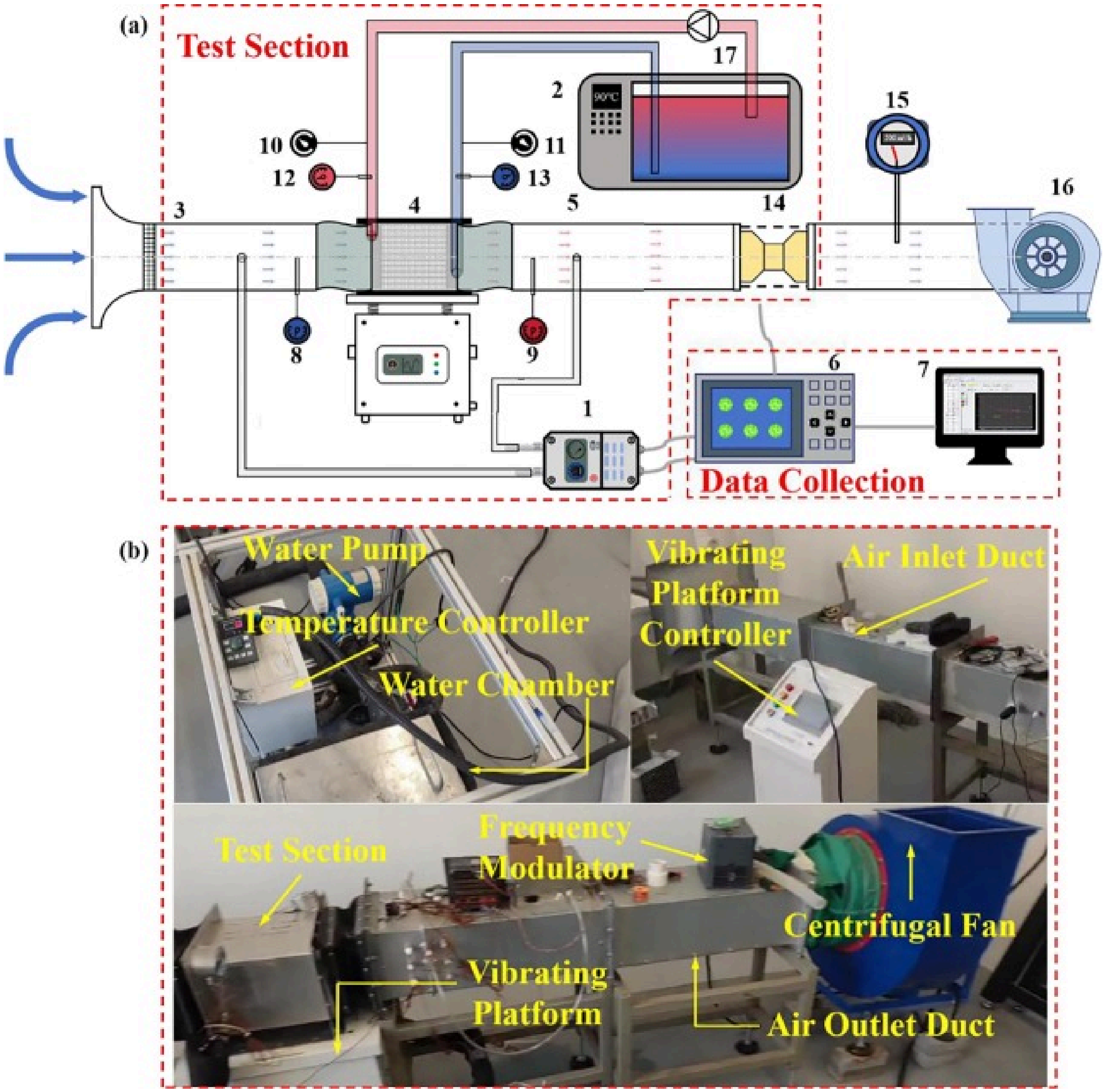

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Fig. 4. Experimental setup (a) schematic diagram; (b) photo of the experimental setup.

Table 3. The main experimental equipment.

1	**Pressure transducer**	**7**	**Computer**	**13**	**Outlet water temperature sensor**
2	Thermostat water bat	8	Inlet temperature sensor	14	Absolute filter
3	Air inlet duct	9	Outlet temperature sensor	15	Air flow meter
4	Test heat exchanger	10	Inlet water pressure sensor	16	Centrifugal fan
5	Air outlet duct	11	Outlet water pressure sensor	17	Water pump
6	Data logger	12	Inlet water temperature sensor		

2.3. Experimental process and data reduction

Before the experiment, calibrate the hot resistance and differential pressure sensors individually. During the experiments, activate the fan and water pump, and set the water flow rate and temperature to the target values. Then, start the vibration platform and set the vibration parameters. After the system runs stably for 30 min, record the temperature and flow rate on both sides and calculate the heat balance error. Data collection is carried out when the heat balance error is less than or equal to 5%. Meanwhile, the vibration platform is turned on. Each test condition is repeated three times with a 5-minute interval between each test, and the average value is taken as the final experimental results.

2.3.1. Acquisition of vibration parameters

In theory, a vehicle has six degrees of freedom regarding displacement-namely, vertical (up and down), longitudinal (front and back), and lateral (left and right) movements [46]. However, empirical evidence shows that vertical displacement and pitch angle significantly impact vehicle dynamics. Thus, this work simplifies the model by focusing on the vertical displacement and pitch angle, particularly analyzing the performance of heat exchanger under the conditions and disregarding other forms of displacement and angular variations. Based on the structural of vehicles, a half-car vibration model was established with the following assumptions: 1) the elasticity of vehicle's components is minimal, and the vehicle is considered a rigid structure; 2) the vehicle is symmetrical, with both sides experiencing identical road conditions and no yaw motion, and each component's mass is concentrated at its center of mass; 3) the effects of engine and other excitations are disregarded, focusing solely on road-induced vibrations. The road surface is treated as a steady random vibration source; 4) the tires stay in contact with the ground, and tire damping is negligible. Based on the assumptions, the vehicle is simplified into a five-degree-of-freedom vibration system model, as shown in Fig. 5.

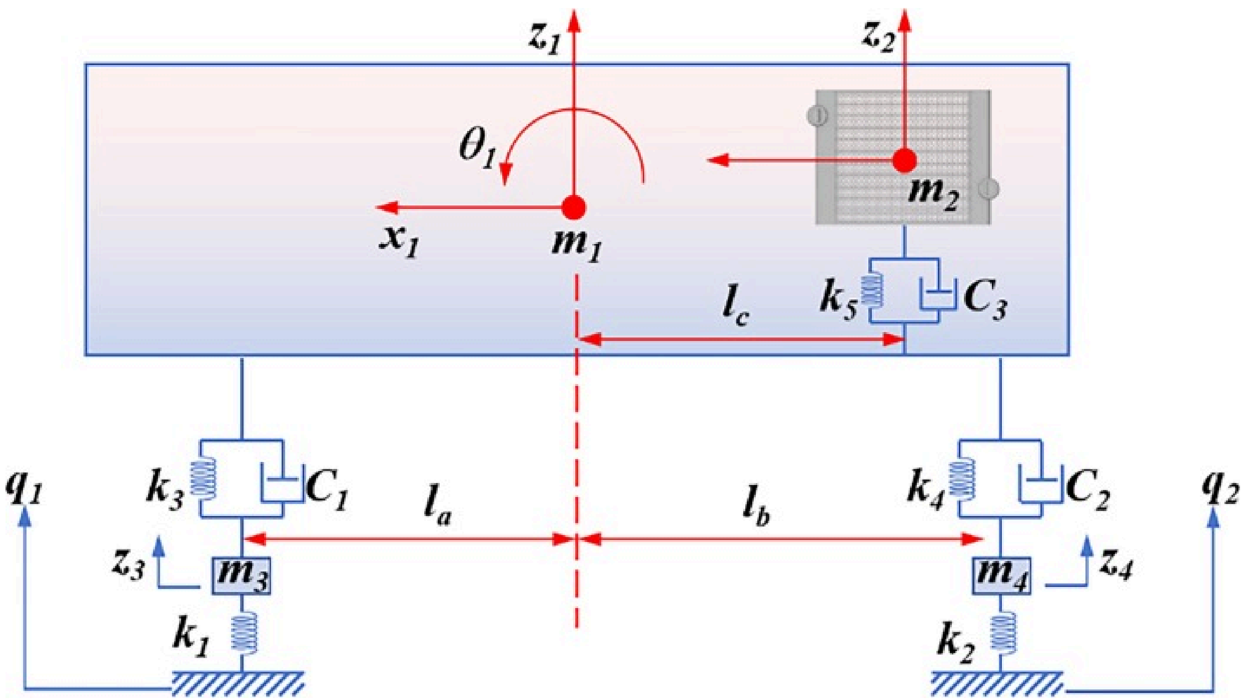

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Fig. 5. Simplified diagram of the vehicle vibration model.

Where, m_1 (kg) is the mass of the vehicle body, m_2 (kg) is the mass of the metal foam heat exchanger, and m_3 (kg) and m_4 (kg) are the masses of the front and rear axles; θ_1 (rad/s) is the pitch angle displacement of the whole vehicle around its center of mass; l_a is the distance from the front axle to the vehicle's center of mass, l_b is the distance from the rear axle to the vehicle's center of mass, and l_c is the distance from the metal foam heat exchanger to vehicle's center of mass, in meters; c_1 (N·s/m) and c_2 (N·s/m) are the damping coefficients of the front and rear suspensions, and c_3 is the damping coefficient of the metal foam heat exchanger support; k_1 (N/m) and k_2 (N/m) are the stiffness of front and rear tires, k_3 (N/m) and k_4 (N/m) are the stiffness of front and rear suspensions, and k_5 is the stiffness of heat exchanger support; q_1 (m)and q_2 (m)are the road excitation displacements of front and rear wheels. Using the Lagrange equation to establish the kinematic equation of vibration system, and organize it into matrix form [47]. The vehicle parameters in the Ref [48,49] were used for calculation.

$$[M]\{\ddot{z}\}+[C]\{\dot{z}\}+[K]\{z\}=\{r\} \tag{1}$$

$$\{r\}=[\Phi]\{q\} \tag{2}$$

Solving the heat exchanger vibration system in frequency domain, applying Fourier transform to both sides of Eq. (1) gives:

$$\left(-4\pi^2 f^2 [M]+j2\pi f [C]+[K]\right)\{z(f)\}=[\Phi]\{q(f)\} \tag{3}$$

In Eq. (3): $\{z(f)\}$ is the Fourier transform of the displacement vector $\{z\}$; $\{q(f)\}$ is the Fourier transform of the road excitation vector $\{q\}$. $\{z\}$ and $\{q\}$ are shown in Eq. (4).

$$\{z\}=\{z_1,\theta_1,z_2,z_3,z_4\}^T \tag{4}$$

$$\{q\}=\{q_1,q_2\}^T$$

The automotive system's frequency response is the ratio of the system displacement vector's Fourier transform to the road excitation vector [50], as shown in Eq. (5).

$$[H(f)] = \frac{\{z(f)\}}{\{q(f)\}} = \left([K] - 4\pi^2 f^2 [M] + j2\pi f [C]\right)^{-1} [\Phi] \tag{5}$$

Road excitation is the input of the road to vehicle vibration system, a function of time, described by the road roughness model, and it is also related to the speed characteristics of the vehicle [51]. The relationship between the spatial power spectral density of road roughness and the temporal frequency spectral density is as follows:

$$G_q(f) = \frac{1}{v} G_q(n) = \frac{1}{v} G_q(n_0) \left(\frac{n}{n_0}\right)^{-w} \tag{6}$$

Where $G_q(f)$ is the displacement spectral density of the road excitation $q(t)$ in the time frequency domain, with units of m^2/m^{-1}; $G_q(n)$ is the spatial power spectral density of road roughness, in m^2/m^{-1}; $G_q(n_0)$ is the value of road roughness spatial power spectral density at the reference spatial frequency n_0 ($0.1\,m^{-1}$), also known as the road roughness coefficient, in m^2/m^{-1}; w is the frequency exponent, generally equal to 2; n is the spatial frequency, in units of m^{-1}, which is the inverse of the wavelength λ, representing how many wavelengths are contained per meter of length. Vehicle speed also affects the input information of the vibration system. When the vehicle travels at speed v over road roughness with spatial frequency n, the tires encounter an excitation frequency of nv. In this work, a Class B road was selected, with a roughness coefficient of $64\times10^{-6} m^2/m^{-1}$ and a range of n from 0.011 to 2.830.

Assuming all wheels follow the same track, the road excitation for each wheel is related as follows:

$$q_i = q_1(t-\tau), \tau_i = \frac{L_i}{u}, i = 1, 2$$

$$q_i(\omega) = q_1(\omega) e^{-j\omega\tau_i} \tag{7}$$

Where L_i is the distance from each wheel to the front wheel; u is the vehicle speed; τis the time lag of each wheel relative to the front wheel. Based on random vibration theory [52], the vibration response's power spectral density can be derived from its frequency response characteristics.

$$G_{z_2}(f) = |H_{z_2-q}(f)|^2 G_q(f) \tag{8}$$

Where $G_{z2}(f)$ is the vertical displacement power spectral density of heat exchanger under road excitation; $H_{z2\text{-}q}(f)$ is the frequency response of heat exchanger in the vertical direction to road excitation. Using MATLAB to calculate the random road excitation of vehicle at a speed of v=60 km/h according to the method described above.

Fig. 6(a) presents the power spectral density plot for the heat exchanger's center of mass acceleration, indicating that road-induced vibration frequencies mainly occur between 20 Hz and 40 Hz. Therefore, this work focuses on this frequency range. Fig. 6(b) presents the calculated vertical displacement due to the heat exchanger's vibration, with amplitudes ranging from 1 mm to 5 mm.

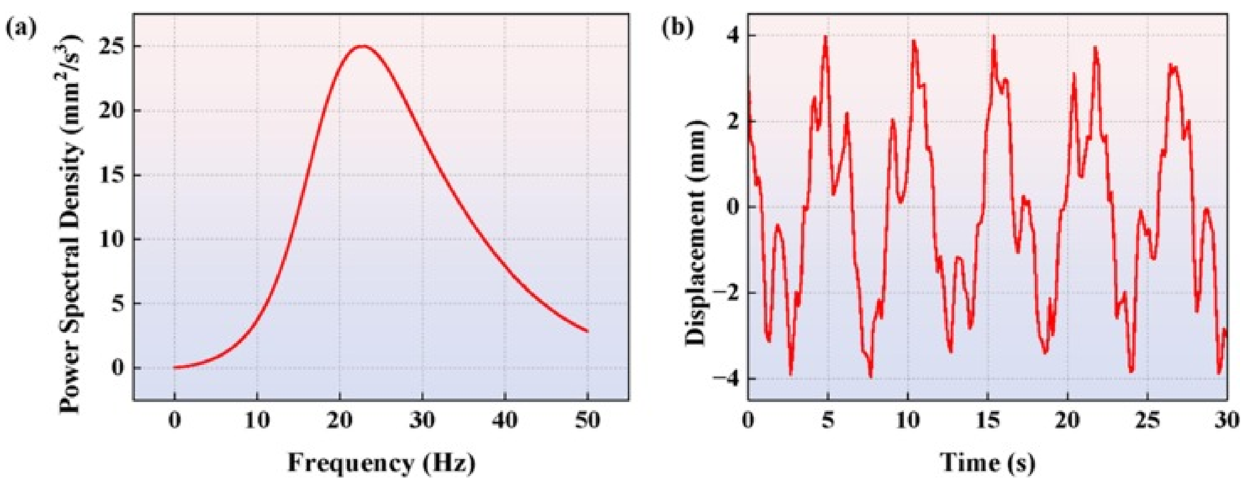

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Fig. 6. Analytical solution of vibration of vehicle heat exchanger: (a) power spectral density of heat exchanger; (b) vertical displacement of heat exchanger.

To study the effect of vibration on the performance of Kelvin metal foam heat exchanger, experiments under vibration and non-vibration conditions were conducted. During the experiments, the water flow rate is maintained at 0.75 m^3/h. The wind tunnel's inlet temperature is adjusted to room temperature, and the water tank is set between 323.15 K and 353.15 K. The inlet pressure remains at atmospheric levels to simulate automotive radiator conditions. Wind speed, adjustable from 1 to 5 m/s via a fan frequency modulator, is chosen based on test platform adaptability.

2.3.2. Data reduction

To accurately evaluate the effect of vibration on performance of metal foam heat exchanger, it's essential to choose suitable performance index [53]. This paper uses Colburn J factor for heat transfer, F-factor for friction, and comprehensive heat transfer factor η to assess its overall performance.

The radiative heat transfer is deemed negligible as the operating temperature is below 400 K [54]. $EHTC$ is determined by Eq. (9), where A_{base} refers to the macroscopic area carrying Kelvin lattice in the heat exchanger, specifically the air side base area listed in Table 1.

$$EHTC = \frac{Q_{total}}{A_{base} \bullet \Delta T_{ml}} \tag{9}$$

Q_{total} is the average of heat dissipation on the water side (Q_w) and heat absorption on the air side (Q_{air}), as indicated in Eq. (10). ΔT_{ml} is the logarithmic mean temperature difference between the wall and air temperatures, defined in Eq. (11).

$$Q_w = \rho_w V_w C_{p,w} (T_{w,in} - T_{w,out}) \tag{10}$$

$$Q_{air} = \rho_{air} V_{air} C_{p,air} (T_{air,out} - T_{air,in})$$

$$Q_{total} = \frac{Q_w + Q_{air}}{2}$$

$$\Delta T_{ml} = \frac{[(T_{w,in} - T_{air,in}) - (T_{w,out} - T_{air,out})]}{ln\left(\frac{T_{w,in} - T_{air,in}}{T_{w,out} - T_{air,out}}\right)} \tag{11}$$

Where, ρ and V denote density and volumetric flow rate, respectively. The subscripts *w* and *air* refer to the working fluids on the hot and cold sides of the heat exchanger, while *in* and *out* indicate the fluid's inlet and outlet. The heat balance error in the experiment is given by Eq. (12).

$$\varphi = 100 \times \left| \frac{Q_W - Q_{air}}{Q_{air}} \right| \tag{12}$$

To assess the kelvin metal foam heat exchanger' performance under vibration, dimensionless numbers were used for statistical analysis. Thermal behavior is mainly affected by heat transfer height rather than pore structure, so air-side height H was used for calculating overall Nu. Definitions for Nu and Re are provided in Eq. (13) and Eq. (14).

$$Nu = \frac{HTC \bullet H}{\lambda_{air}} \tag{13}$$

$$Re = \frac{\rho_{air} v H}{\nu}, \; v = \frac{V_{air}}{A_{flow}} \tag{14}$$

The fluid velocity in metal foam heat exchanger is v, and A_{flow} is the actual flow area, assuming fully developed flow. The performance of the heat exchanger is evaluated using the parameter $J/F^{1/3}$, called "volume goodness factor (η)," [55] where J is the Colburn factor (Eq. (15) and F is the friction factor (Eq. (16). Given that the Kelvin metal foam predominantly contribute to flow losses, the hydraulic diameter, as defined by Eq. (17), is used to determine the friction factor.

$$J = \frac{Nu \bullet Pr^{1/3}}{Re} \tag{15}$$

$$F = \frac{2D_h}{\rho_{air} v^2} \bullet \frac{\Delta P}{\Delta L} \tag{16}$$

$$D_h = 4\frac{\varepsilon}{\alpha_{sf}} \tag{17}$$

Where, ΔP is the pressure loss across the metal foam heat exchanger.

2.3.3. Uncertainty

During the experiments, the inlet and outlet temperatures, the pressures on both sides, and the air flow rate were directly measured. Table 4 lists the main parameters and performance specifications of the measuring equipment.

Table 4. Specifications of the equipment used in the experiment.

Instrument	Accuracy	Range
WRNT-191 thermistor	±0.3%FS	−233.15 K ~ 398.15 K
HD130 pressure difference transmitter	0.25%FS	−0 ~ 2 kPa
MS6252B hot-wire anemometer	0.01 m/s	0.4 ~ 30 m/s
Y100L4 centrifugal fan	/	0～6640 m^3/h
EV4300-3G3 frequency transformer	/	0 ~ 400 Hz
PFT-LD-32 water flowmeter	±0.5%FS	0 ~ 35 m^3/h
Water pump	/	2900r/min
Z-6-S Vibration table	/	1 ~ 600 Hz; 0 ~ 5 mm

The indirectly measured parameters include heat transfer amount Q, equivalent heat transfer coefficient $EHTC$, J factor, f factor, η factor. According to the principles of error analysis [56], the indirectly measured value Y can be expressed as a function of several independent variables $f(x, y \ldots z)$. The transfer function for the combined uncertainty is delineated in Eq. (18), while the uncertainty associated with the indirect measurement is defined in Eq. (19).

$$u_c(Y)=\sqrt{\left[\frac{\partial f}{\partial x}u_c(x)\right]^2+\left[\frac{\partial f}{\partial y}u_c(y)\right]^2+\cdots+\left[\frac{\partial f}{\partial z}u_c(z)\right]^2} \tag{18}$$

$$U=\left|\frac{u_c(Y)}{Y}\right|\times 100\% \tag{19}$$

Where, partial derivative term represents the sensitivity coefficient. $u_c(x)$, $u_c(y)$…$u_c(z)$ represent the measurement errors. Table 5 lists the maximum uncertainties for each measurement.

Table 5. Maximum uncertainty of Measured Values.

Parameter	**Maximum error value**	**Parameter**	**Maximum error value**
Temperature (K)	±0.5K	*F* factor	±6.5%
Pressure (Pa)	±5 Pa	*J* factor	±5.8%
Q_{total} (W)	±3.3%	η factor	±7.3%
EHTC (W/m^2·K)	±5.3%	ΔT_{ml}	±6.9%

2.4. Simulation model and method

This work also uses numerical simulations to study how vibration affects forced convection heat transfer in the metal foam heat exchanger, aiming to understand the mechanisms behind their improved heat transfer properties.

2.4.1. Governing equations and assumptions

The minimum *Re* number from Eq. (14) is 279. According to Ref [57], *Re* numbers above 300 indicate fully turbulent flow in metal foam. Therefore, choosing the right turbulence model is crucial for accurate simulation analysis. The *k*-ω model is commonly used to address issues involving bounded walls and low *Re* numbers, particularly in scenarios involving flow separation and transition. The *k*- ω model heavily depends on the wall mesh, requiring the boundary layer mesh to be closer to the wall for accuracy. This led to a boundary layer mesh mass of less than 0.2 during CHT calculations, negatively impacting heat transfer calculations. Therefore, this work does not use *k*- ω turbulence model for simulations. Following Sun et al. [58], *k*- ε model as used for its efficiency and accuracy. Set Y^+ equal to 30 under this model while using the enhanced wall function to maintain accurate calculations of the viscous sublayer [59].

- Continuity equation:

$$\frac{\partial(\rho_f u_i)}{\partial x_i} = 0 \tag{20}$$

- Momentum equation:

$$\frac{\partial(\rho_f u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_j} + \frac{\partial}{\partial x_j}(\mu_f + \mu_t)\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{21}$$

- Energy equation for fluid:

$$\frac{\partial(\rho_f C_{pf} \mu_j T)}{\partial x_j} = \frac{\partial}{\partial x_j}\left(\lambda_f \frac{\partial T_f}{\partial x_j}\right) \tag{22}$$

- Turbulent kinetic energy: *k*

$$\frac{\partial \rho k}{\partial t} + \frac{\partial \rho u_j k}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] + \mu_t \frac{\partial u_i}{\partial x_j}\left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j}\right) - \rho\varepsilon \tag{23}$$

- Turbulent kinetic energy dissipation:ε

$$\frac{\partial \rho \varepsilon}{\partial t} + \frac{\partial \rho u_j \varepsilon}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right] + \frac{c_1 \varepsilon}{k}\mu_t \frac{\partial u_i}{\partial x_j}\left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j}\right) - c_2 \rho \frac{\varepsilon^2}{k} \tag{24}$$

The simulations were conducted using ANSYS Fluent, selecting the SIMPLE method for pressure–velocity coupling due to its superior handling of complex boundaries and faster convergence. The enhanced wall function method was applied to address heat transfer on the secondary surface. A second-order method was used for pressure discretization to enhance accuracy, while second-order upwind schemes were used to the momentum and energy equations. The default relaxation factor and gravity at $9.81\,m/s^2$ were maintained. Heat transfer primarily occurs on the secondary surface, so the enhanced wall function method was used for precise wall heat transfer capture. Convergence was evaluated using residuals, with momentum and continuity set to 10^{-5} and energy to 10^{-6} for higher accuracy. To simulate CHT, five simplification are applied: 1) Air is incompressible; 2) Solids and liquids have constant thermal properties; 3) Radiative heat transfer is omitted; (4) Wall friction is neglected; 5) Dissipative heat from viscous forces is ignored.

2.4.2. Models and boundary conditions

The simulations aim to study vibration-enhanced heat transfer in Kelvin metal foam, necessitating applying vibration to the Kelvin wall. In Fluent, component vibration is typically implemented using dynamic mesh technology and overset mesh technology. The dynamic mesh model includes the smoothing, local remeshing, and dynamic layering models. The fairing model is divided into the spring and diffusion fairing methods, which are widely used but not ideal for large deformations and are computationally intensive. The moving boundary layer method offers better accuracy but requires a hexahedral mesh. The local grid redrawing model works with both structured and unstructured grids. In practice, the complexity of 3D models often causes calculation divergence due to negative volumes, making the overlap grid method preferable for mobile grid settings.

Overlap grid technology [60] consists of two separate grids: a background grid and a component grid. In overlapping areas, physical quantities are interpolated. After dividing the background grid globally, the component grid is further subdivided. A key benefit of this technology is that it allows the component grid to move freely within the background grid, which is particularly useful for solving dynamic heat transfer problems [61].

The geometry model is shown in Fig. 7(a). Considering the heat exchanger's symmetrical structure and the periodic arrangement of its fins along the hot fluid flow, a diaphragm and Kelvin metal foam unit fins were selected for study. The computational domain has been extended to mitigate the effects of inlet disturbances and outlet backflow. During the simulation, the inlet is configured as a velocity inlet at 300K, and the outlet as a pressure outlet. Due to the thinness of the baffle, heat transfer loss to the baffle is ignored. The lower baffles of the Kelvin metal foam are set with a first-type boundary condition at 323.15K~353.15K, while the Kelvin metal foam's other surface in contact with air uses a coupling thermal boundary condition. Overlap grid technology and UDF are employed to define the vibration of Kelvin foam [62]. The motion equation is described in Eq. (25).

$$V = 2\pi f_v A_v \cos(2\pi f_v t) \times 10^{-3} \quad (25)$$

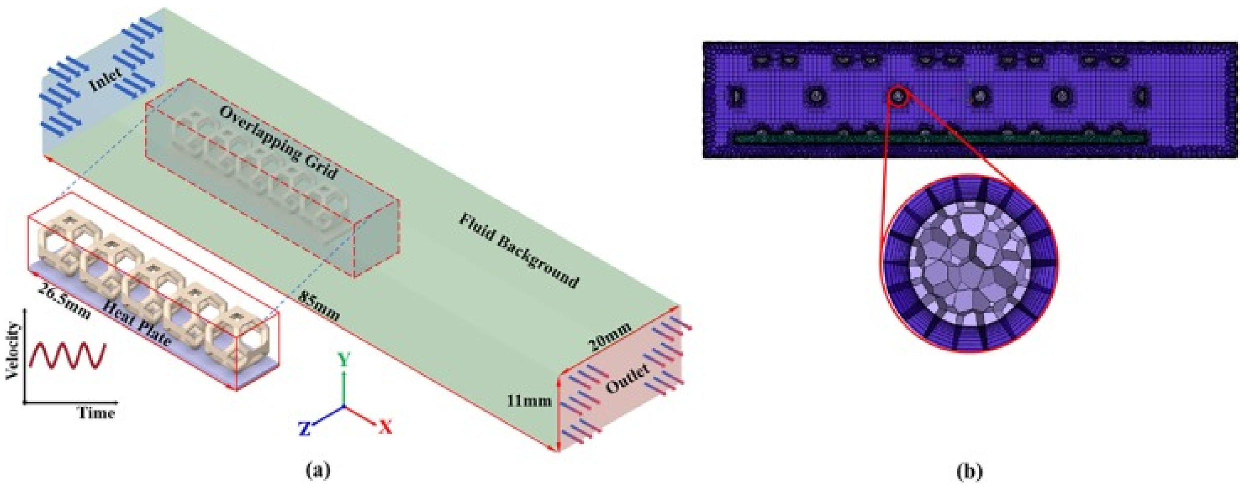

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Fig. 7. Vibration conjugate heat transfer calculation model of metal foam (a) model and boundary conditions; (b) mesh and the detail of boundary layer.

2.4.3. Mesh generation and grid independence verification

The computational model was meshed using Fluent Meshing software. The grid generation for various structures depends on factors like geometric complexity, desired physical parameters, and computational resources. Due to the complex Kelvin metal foam structure, a pure hexahedral mesh was inadequate for achieving congruence between fluid and solid domain meshes. Thus, an unstructured tetrahedral mesh was used, with a finer boundary layer mesh at the solid–fluid interface. This method is common for simulating metal foam and complex porous structures. The grid details are shown in Fig. 7(b).

To ensure accurate calculations, the model's grid independence is verified. Table 6 shows results from tests with four grid sizes at air inlet speeds of 2 m/s and 5 m/s under 40 Hz-5 mm vibration conditions. The temperature difference between the inlet and outlet is used as the verification index. An error below 2% with the largest grid size confirms accuracy. Finally, a grid of 1.05 million was chosen for calculating the vibrational conjugate heat transfer of the Kelvin metal foam.

Table 6. Temperatures differences and *EHTC* between inlet and outlet at 2 m/s and 5 m/s.

Total Elements (10^6)	**Inlet 2 m/s**		**Inlet 5 m/s**	
	ΔT (K)	**Difference**	***EHTC* (W/m^2·K)**	**Difference**
0.45	1.77	11.85%	3.05	8.16%
0.76	1.51	9.63%	2.93	3.90%
1.05	1.37	1.48%	2.84	0.71%
1.33	1.37	1.48%	2.82	0.35%
1.85	1.35	0.00%	2.82	0.00%

3. Result and discussion

This section discusses the experimental process and simulation results. Firstly, the reliability of the heat balance in a forced convection heat transfer experiment under vibration is verified. Next, how different vibration parameters affect heat transfer performance of the metal foam heat exchanger is analyzed. Finally, the mechanism by which vibration enhances heat transfer is examined using conjugate heat transfer simulation results.

3.1. Balance of heat transfer between hot side and cold side

In theory, the heat absorbed on the cold side of the heat exchanger should equate to the heat transferred on the hot side. However, discrepancies between these values may arise due to thermal conductivity resistance and measurement errors. The experimental boundary conditions are detailed in Section 2.3.1. Notably, the vibration heat transfer experiment involving the Kelvin metal foam heat exchanger primarily considers an inlet air velocity of 5 m/s and a water-side temperature of 353.15 K. Fig. 8 reveals that the thermal balance error remains within ±5%, indicating the reliability of the experimental data.

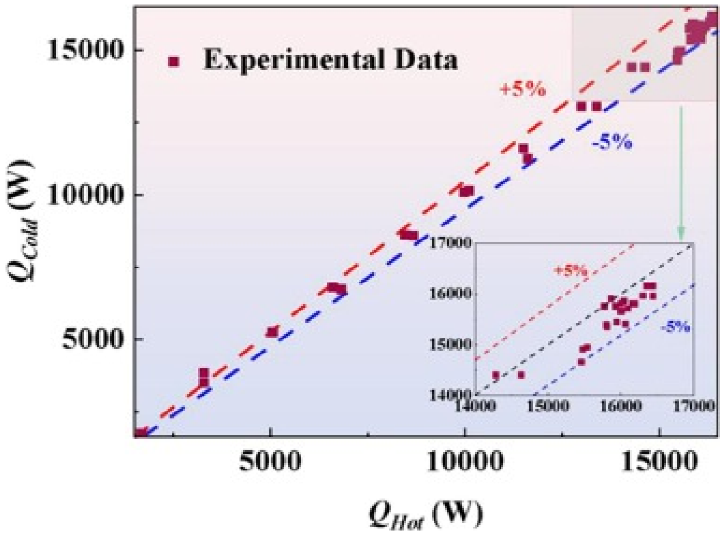

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Fig. 8. Heat balance validation between the cold stream and hot stream.

3.2. Experimental results

As illustrated in Fig. 9(a), the *EHTC* of Kelvin metal foam heat exchanger exhibits an increase with increasing air flow rates under vibrational conditions, mirroring the growth trend observed under non-vibrational conditions. Notably, variations in the heat side temperature do not influence the *EHTC* of the heat exchanger. Take the vibration condition of 40 Hz–4 mm as an example, at an air flow rate of 1 m/s, the *EHTC* is measured at 333.69 W/m^2·K, whereas at an air flow rate of 5 m/s, it rises to 1520.64 W/m^2·K. Increasing air velocity boosts the *EHTC* by approximately 361.35%. Kelvin metal foam with a porosity of 0.85 exhibits significantly higher flow resistance. At air flow rates of 1 m/s and 5 m/s, the pressure drop per unit length is 1.1 kPa/m and 22 kPa/m, respectively. Additionally, there is no significant change in pressure difference across the heat exchanger under vibration and non-vibration conditions, indicating that Kelvin metal foam's flow resistance remains unaffected.

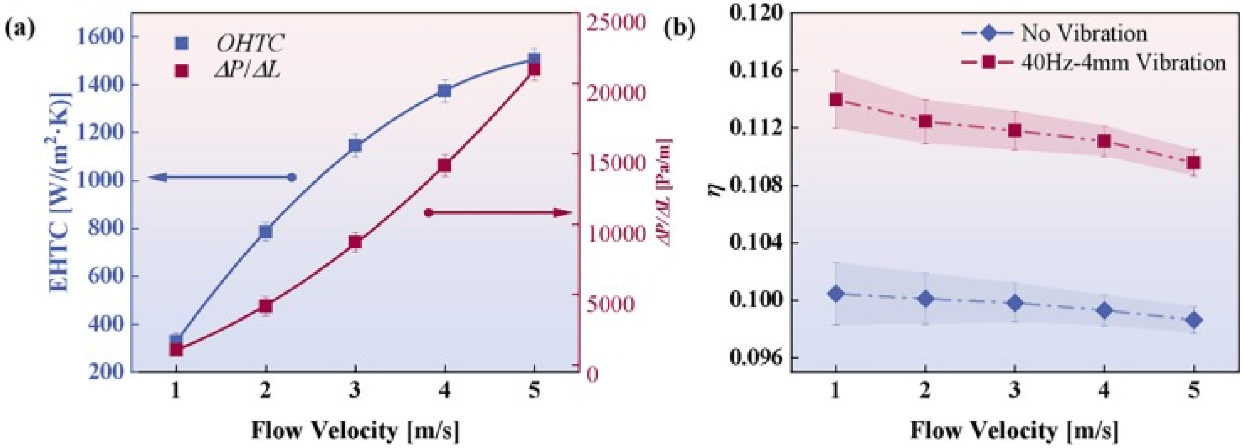

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Fig. 9. Thermal hydraulic performance changes with air velocity and heat source temperature.

Fig. 9(b) illustrates the comprehensive heat transfer factor η of Kelvin metal foam at a vibration state of 40Hz-4mm. As wind speed increases, η declines due to rising flow resistance, despite higher flow rates enhancing heat transfer. However, this increase in heat transfer is slower than the rise in flow resistance. Notably, the overall performance of the heat exchanger is markedly improved under vibrational conditions compared to non-vibrational conditions. At an air flow velocity of 1 m/s, η are 0.114 and 0.100 for the vibrational and non-vibrational conditions, respectively. At an air flow velocity of 5 m/s, these factors are 0.110 and 0.099, respectively. On average, vibration results in an 11.2% increase in the overall heat transfer factor of the Kelvin metal foam heat exchanger.

3.3. Result analysis

This section quantitatively analyzes the impact of vibration on the thermal performance of the Kelvin metal foam heat exchanger using the control variable method. It examines vibration test results and the effects of different vibration conditions, concentrating on

EHTC and overall heat transfer factors since vibration has minimal impact on flow resistance.

3.3.1. Equivalent heat transfer coefficient

The *EHTC* of the Kelvin metal foam heat exchanger is determined by measuring the temperature differential between the inlet and outlet on both cold and hot sides, utilizing Eq. (9), (11), shown in Fig. 10(a). To investigate the effect of vibration on heat transfer performance of the Kelvin foam heat exchanger, the experimental conditions were set with an inlet air velocity of 5 m/s and a water-side temperature of 353.15 K. The results indicate that *EHTC* increases with both frequency and amplitude of the vibration. Without vibration, the *EHTC* is 1366.11 W/m^2·K. At a vibration frequency of 20 Hz and amplitude of 1 mm, *EHTC* slightly increases to 1370.11 W/m^2·K. However, with frequency and amplitude raised to 40 Hz-5 mm, *EHTC* significantly rises to 1556.37 W/m^2·K. Fig. 10(b) illustrates *EHTC* growth rates under various vibrations, showing that *EHTC* increases with amplitude at the same frequency. At 20 Hz, the growth rate is minimal, between 0.29% and 4.24%. At 40 Hz, *EHTC* shows the highest growth rate with amplitude, between 3.83% and 13.93%. *EHTC* rises with increasing vibration frequency at the same amplitude. At 1 mm amplitude, the growth rate with frequency is lowest, from 0.29% to 3.83%. At 5 mm amplitude, *EHTC* ranges from 4.24% to 13.93%. For the Kelvin metal foam heat exchanger, increasing vibration frequency enhances the heat transfer performance more effectively than increasing amplitude, especially noticeable from 25 Hz onwards.

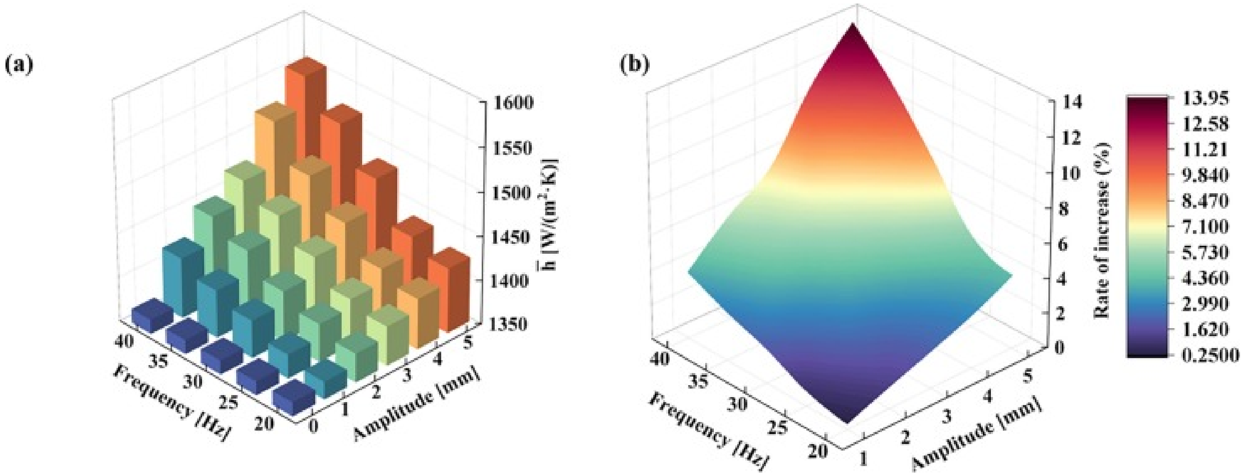

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Fig. 10. (a) *EHTC* of metal foam heat exchanger; (b) growth rate of *EHTC* with vibration parameters.

The Analysis of Variance (ANOVA) is a statistical approach utilized to determine whether there are significant differences in the means among multiple data groups. It is commonly used to evaluate the impact of one or more independent variables on a dependent variable [63]. This is achieved by partitioning the total variability of data into the variability attributable to the independent variables and the residual error [64], as demonstrated in Eq. (26).

$$\text{(a)}\ \ SST = SSR + SSE$$

$$\text{(b)}\ \ SST = \sum_{i=1}^{n}\left(Result_i - Result\right)^2 \quad (26)$$
$$SSR = \sum_{j=1}^{m}\left(Re\,\hat{s}ult_j - Re\bar{s}ult\right)^2$$
$$\mathrm{SSE} = \sum_{i=1}^{n}\left(\mathrm{Result}_i - Re\,\hat{s}ult_i\right)^2$$

$$\text{(c)}\ \ F = \frac{SSR\,/df_{model}}{SSE\,/df_{error}}$$

Where, SST denotes the total variability of the dependent variable in the dataset; *SSR* reflects the variability explained by the independent variable; *SSE* represents variability due to random errors or factors outside the model. For each independent variable, *F*-statistic is computed; a larger *F*-statistic signifies a more substantial influence of the variable on the outcome. Table 7 shows the two-factor ANOVA results, with *F*-values of 29.94 for frequency and 29.74 for amplitude, indicating both have significant and similar effects. The *P*-values for both variables are near zero, confirming their statistically significant impact.

Table 7. ANOVA results of kelvin metal foam heat exchanger vibration heat transfer experiment.

	Sum of squares	d_f	***F***	***P(>F)***
Frequency	26309.84	4	29.94	3.00E-07
Amplitude	26128.74	4	29.73	3.14E-07
Residual	3514.70	16	–	–

3.3.2. Comprehensive performance comparison

To evaluate how vibration affects the performance of metal foam heat exchanger, the comprehensive heat transfer factor η was analyzed under the condition that the inlet air flow rate was 5 m/s. Fig. 11(a) shows the relationship between vibration parameters and η. It is evident that as the amplitude gradually increases, η also increases. The increase rate initially progresses slowly and then accelerates. At an amplitude of 40 Hz, the enhancement effect ranges from 3.85% to 13.86%. Fig. 11(b) illustrates the impact of amplitude on η. As amplitude increases, η also rises, albeit at a consistent growth rate, as evidenced by the stable slope observed in the data. Specifically, when the amplitude reaches 5 mm, the enhancement effect increases from 4.18% to 13.86% with varying frequency. This supports the ANOVA analysis results, showing that both amplitude and frequency positively affect heat transfer of the metal foam heat exchanger.

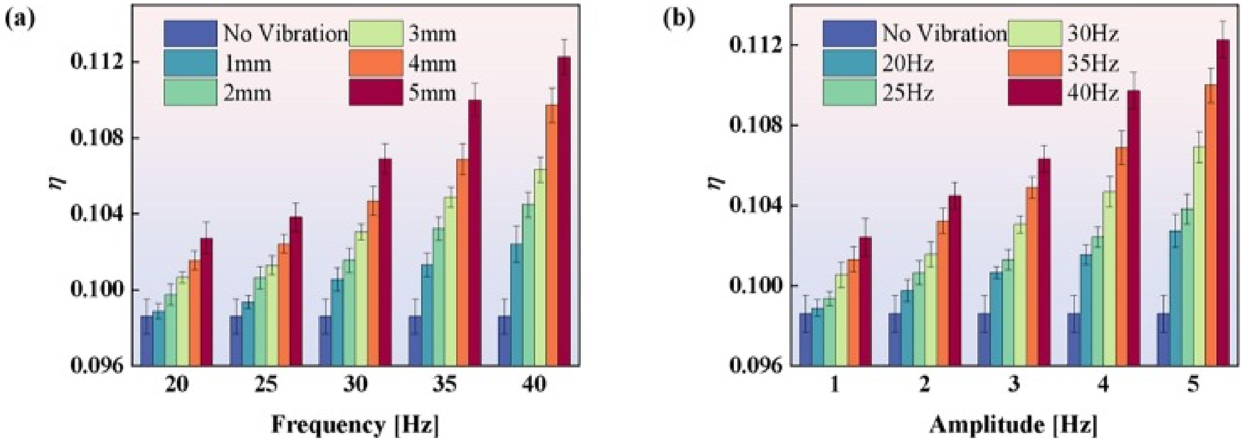

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Fig. 11. Comprehensive performance in different vibration parameters: (a) frequency; (b) amplitude.

3.4. Mechanism of forced convective heat transfer under vibration conditions

The simulations aim to investigate the mechanism by which vibration enhances convective heat transfer of the metal foam heat exchanger. In this regard, vibration parameters are more critical than flow velocity. Therefore, an inlet flow velocity of 5 m/s was selected for simulations.

3.4.1. Data validation

To ensure the reliability of the simulation calculations and subsequent analyses, the volume heat transfer index is used to compare experimental and simulation data. The experiment utilized a full-scale Kelvin metal foam heat exchanger, whereas the simulation focused solely on the vibration of its air side. Consequently, a volume average was used to facilitate dimensional consistency in the comparison, as illustrated in Fig. 12.

The heat dissipation per unit volume differs by±5%, showing consistency between simulation and experimental results, indicating the reliability of the simulation.

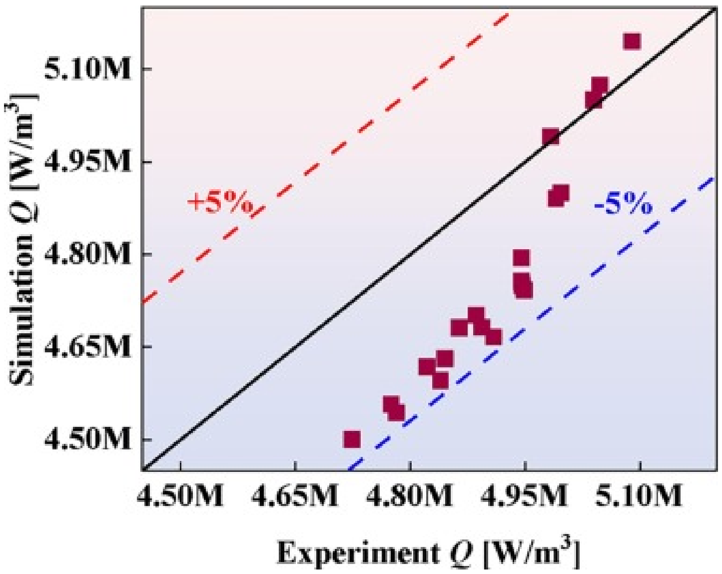

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Fig. 12. The experiment and simulation are compared by unit volume power.

3.4.2. Flow field analysis

Fig. 13 presents a comparative analysis of the Kelvin metal foam's velocity field distribution: the vertical axis shows the distribution during a complete vibration period, and the horizontal axis displays distributions at varying vibration intensities within the same period.

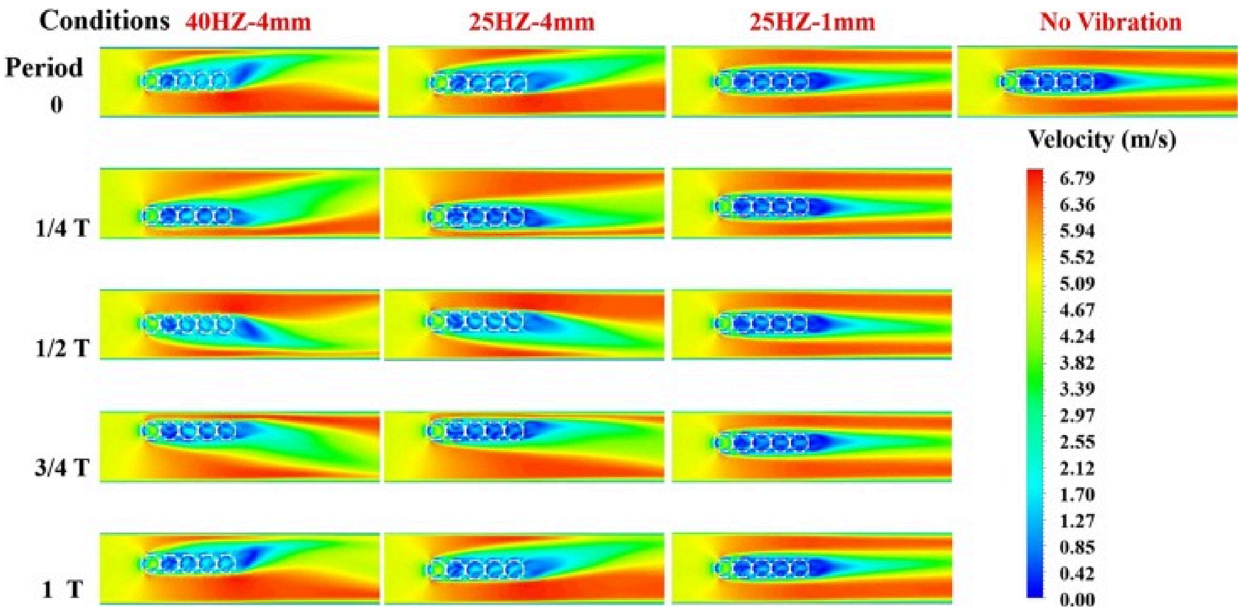

Fig. 13. Distribution of velocity field under vibration conditions.

Under different vibration conditions, the Kelvin metal foam shows higher internal and external flow velocities compared to when there's no vibration. The air velocity distribution within the plane is affected by the variations in vibration, exhibiting a hysteresis effect. A consistent wake turbulence region is also noted at the metal foam's trailing edge. At T/4 and 3T/4, the Kelvin metal foam is at maximum harmonic vibration displacement, with zero velocity and positioned opposite to the motion direction. Consequently, the wake distribution at T/4 contrasts with that at 3T/4, while the velocity distribution remains similar. At 3T/4, the Kelvin metal foam is at the peak displacement with zero acceleration, resulting in a larger flow speed due to the narrow flow range above it. Thus, the air velocity above the Kelvin metal foam exceeds that below. Regarding the macroscopic velocity field distribution, the flow's inertia causes the wake turbulent region to incline, resulting in a pronounced velocity gradient within the mainstream region. At T/2 and T, the Kelvin metal foam reaches the starting point of simple harmonic oscillation with maximum velocity, but the direction of acceleration differs between these two instances. Newton's second law states that the air above the Kelvin metal foam retains its original motion, resulting in lower velocity than the air below it. This phenomenon reverses at time T/2.

The average velocity data is extracted from the air domain within 2 mm of each displacement around the Kelvin metal foam, as shown in Table 8: I. At rest, the air near the Kelvin metal foam moves at an average speed of 2.422 m/s. During one vibration cycle, the average speeds at T/4, T/2, 3T/4, and T are 2.5122 m/s, 2.7430 m/s, 2.5277 m/s, and 2.7403 m/s, respectively. Vibration causes a more irregular velocity distribution and increases overall airflow, which improves convective heat transfer and boosts heat dissipation efficiency. The preceding analysis of the velocity field throughout a complete vibration cycle reveals the following: 1) At vibration periods of T/4 and T, the wind speed adjacent to Kelvin metal foam is observed to be higher; 2) The air speed directly above the Kelvin metal foam is greater at periods of T/2 and 3T/4. An increase in wind speed correlates with an intensification of convective heat transfer.

Table 8. Comparison of average air velocity within 2 mm around Kelvin metal foam.

I	**Period**	**T/4**	**T/2**	**3T/4**	**T**
	Vf (m/s)	**2.5122**	**2.7430**	**2.5277**	**2.7403**
II	Condition	25 Hz, 4 mm	40 Hz, 4 mm	25 Hz,1 mm	No vibration
	V_f (m/s)	2.73075	2.81893	2.45114	2.42188

The effect of different vibration intensities on the flow field is analyzed by comparing horizontal results in Fig. 13. The vibration of the Kelvin metal foam influences the distribution of the flow field. When maintaining a constant vibration amplitude, a comparison between vibration frequencies of 40 Hz and 25 Hz reveals that higher frequency vibrations substantially increase the flow velocity within the Kelvin metal foam compared to lower frequency vibrations. High-frequency vibrations create more wake and significantly disrupt the original flow, increasing local velocity, compared to low-frequency vibrations. An increase in vibration frequency within a certain range leads to greater turbulence. When comparing conditions with the same frequency but different amplitudes, higher amplitudes increase turbulence in the Kelvin metal foam and its wake, disrupting the boundary layer and improving heat transfer. Additionally, the mean air velocity surrounding the Kelvin metal foam has been calculated and is presented in Table 8: II. To summarize, vibration conditions cause a more irregular velocity field, increased turbulence, and higher overall and maximum air velocities compared to rest conditions.

3.4.3. Temperature field analysis

Vertical direction of Fig. 14 illustrates the temperature distribution over a full cycle under various vibration conditions. At 40 Hz–4 mm, vibrating Kelvin metal foam causes more random and symmetrical air temperature changes than when stationary, creating temperature wakes that expand the heating range of the surrounding air. Vibration disrupts the stable boundary layer of air, causing uneven flow and enhancing surface convective heat transfer, leading to a lower temperature in fiber of metal foam compared to non-vibration conditions. This occurs not only because the Kelvin metal foam fiber cuts streamlines but also vibration destroys the viscous sublayer near the wall, increasing heat transfer. During the simulation, the air within 2 mm around the Kelvin metal foam was analyzed for average temperature, as detailed in Table 9: I. Without vibration, the average temperature is 327.37 K. With vibration, temperatures at T/4, T/2, 3T/4, and T are 326.861 K, 325.844 K, 326.87 K, and 325.948 K, respectively, indicating enhanced heat transfer efficiency.

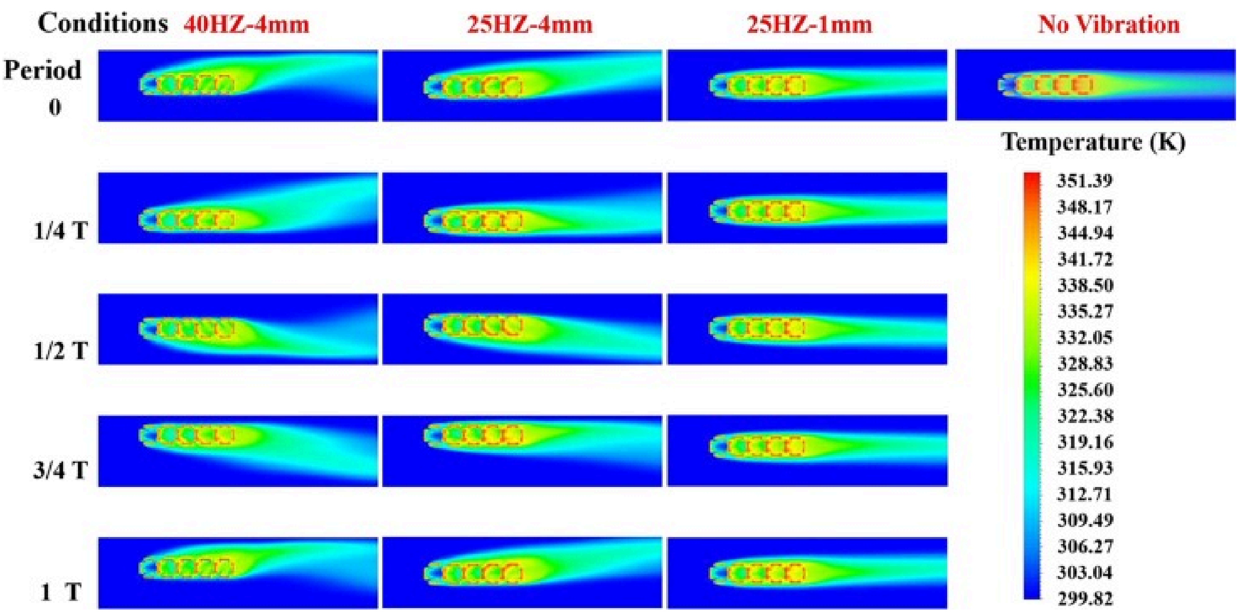

Fig. 14. Distribution of temperature field under vibration conditions.

Table 9. Comparison of average air temperature within 2 mm around the Kelvin metal foam.

I	**Period**	**T/4**	**T/2**	**3T/4**	**T**
	T_f (K)	**326.861**	**325.844**	**326.870**	**325.948**
II	Condition	25 Hz 4 mm	40 Hz, 4 mm	25 Hz,1 mm	No vibration
	T_f (K)	325.645	325.161	327.211	327.372

The transverse comparison results show that increasing frequency significantly reduces the metal foam's temperature, even with a constant amplitude. This phenomenon corresponds to the velocity field, where an increase in frequency further extends the wake region. At a constant frequency, higher amplitude shear enhances the interaction with the boundary layer of the stable flow field, resulting in a slightly lower temperature of the metal foam compared to conditions with lower amplitude. Both frequency and amplitude positively influence heat transfer. Air within 2mm around Kelvin metal foam is also selected to calculate the average temperature, as shown in Table 9: II.

3.4.4. Quantitative analysis within a vibration period

The heat transfer coefficient from simulation is calculated by Eq. (27). The simulation-derived h_{sim} accounts for the heat flux at the surface of the heat transfer cell, the average temperature of the solid, and the fluid temperature across the region. Consequently, a significant divergence arises between h_{sim} from the simulation and *EHTC* for the entire heat exchanger. This variance is justifiable, given that the analysis is conducted under consistent evaluation criteria.

$$h_{sim} = \frac{q}{T_s - T_f} \tag{27}$$

Fig. 15 (a) and (b) illustrate the simulated variation of the heat transfer coefficient under vibration. It shows that the heat transfer coefficient fluctuates with time due to vibration. For a 25Hz frequency with varying amplitudes, the average heat transfer coefficients during a simple harmonic period are 95.13W/(m^2·K), 96.45W/(m^2·K), 98.03W/(m^2·K), 102.40W/(m^2·K), and 104.56W/(m^2·K). In contrast, the stationary Kelvin metal foam has a mean heat transfer coefficient of 94.32W/(m^2·K). In comparison to the static condition, the average heat transfer coefficient exhibited an increase of 10.85% under vibrational conditions. Specifically, under a vibration frequency of 40Hz, the average heat transfer coefficients were measured to be 96.75W/(m^2·K), 99.78W/(m^2·K), 103.39W/(m^2·K), 107.15W/(m^2·K), and 110.82W/(m^2·K), respectively. The maximum observed increase in the average heat transfer coefficient was 17.49%.

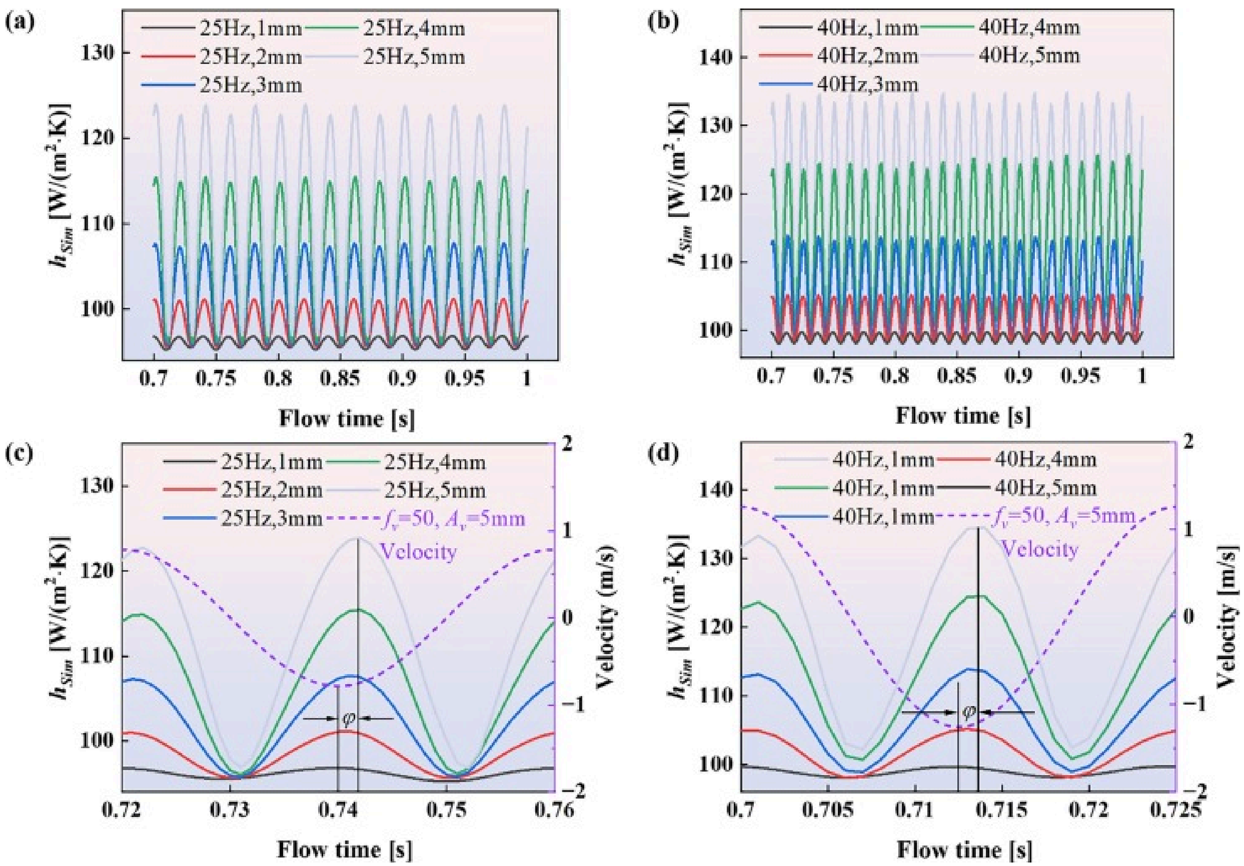

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Fig. 15. Quantitative analysis of heat transfer coefficient variation with vibration: (a) different amplitudes at 25Hz within 0.7s~1.0s; (b) different amplitudes at 40Hz within the time range of 0.7s to 1.0s; (c) different amplitudes of 25Hz; (d) different amplitudes at 40Hz.

In a single cycle, as illustrated in Fig. 15(c) and (d), it is evident that during the vibration period of the Kelvin metal foam, the velocity exhibits two distinct changes (as indicated by the dashed line). The velocity variation causes the heat transfer coefficient to change periodically, with a period half that of the vibration. In the simulation, the Kelvin metal foam starts at the air domain's center. In the simulation, the initial position of the Kelvin metal foam is set at the center of the air domain. Throughout the subsequent motion, the maximum velocity is consistently attained upon passing through the central position, leading to the convective heat transfer coefficient approaching its maximum value at the central position. Since the Kelvin metal foam's velocity is zero at T/4, the heat transfer coefficient reaches its minimum. This indicates that the heat transfer coefficient's variation frequency is twice that of the vibration frequency. Fig. 15 illustrates that there is consistently a phase lag, denoted as angle φ, at the extremum of the heat transfer coefficient. This phenomenon is attributed to the hysteresis effect of air movement. The amplitude ranges from 1 mm to 5 mm, and the hysteresis angle increases progressively with the amplitude.

Fig. 16 illustrates the pressure differential obtained before and after air flow through the Kelvin metal foam over a specified period. Evidently, the pressure differential at the simulated inlet and outlet varies in conjunction with the Kelvin metal foam’s vibration. This vibration is associated with an increase in flow resistance, attributable to the disturbance it introduces to the airflow. The resultant increase in turbulence consequently elevates the flow resistance. Under vibration conditions of 25 Hz–5 mm and 40 Hz–5 mm, the maximum downstream resistances are 424.53 Pa/m and 622.64 Pa/m, respectively. This value is small and may be negligible in thinner heat exchanger; however, it should be considered as a variable in the design of a large heat exchanger.

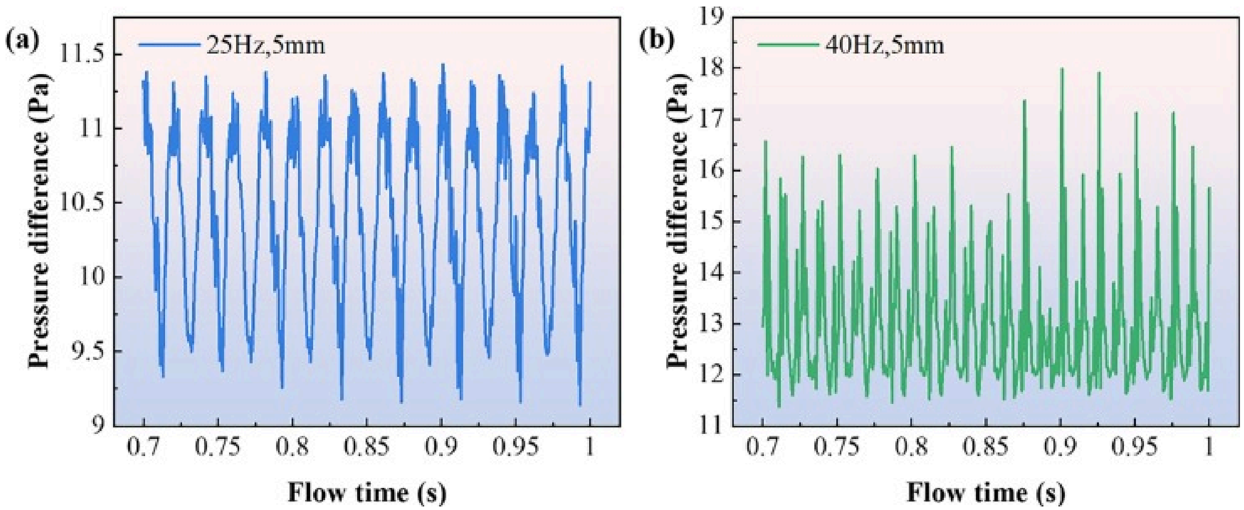

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Fig. 16. Flow resistance of the Kelvin metal foam within a cycle (a) 25Hz-5mm; 40Hz-5 mm.

3.4.5. Evaluation of vibration-enhanced heat transfer

The comprehensive performance evaluation factors are necessary for a macroscopic evaluation of the vibration-induced heat transfer enhancement in Kelvin metal foam heat exchanger. The evaluation factors, typically used in heat exchanger as outlined in Section 2.3.2, were utilized. The simulations ware conducted under the conditions: an inlet flow rate of 5m/s, a temperature of 300K, and a heat source temperature of 353.15K.

Fig. 17(a) illustrates the distribution of *J*-factor obtained via simulations under different vibration conditions. The *J*-factor increases with higher frequency and amplitude, which aligns with previous experimental results. At a vibration of 40Hz-1mm, the *J*-factor is 0.0125, and at 40Hz-5mm, it is 0.0136, marking increases of 4.6% and 14.3% from the non-vibration *J*-factor of 0.012. Fig. 17(b) illustrates the variation of friction factor *F* with frequency. With the increase of vibration frequency, *F* exhibits an upward trend, with a maximum increase of 11.92%. This peak increase occurs at a frequency of 40 Hz and amplitude of 5 mm. Conversely, under conditions of minimal amplitude of 1 mm, F increases by only 0.3%. Therefore, it's crucial to consider the increased flow resistance caused by vibration when Kelvin foam metal heat exchanger operate under high amplitude and frequency for long durations.

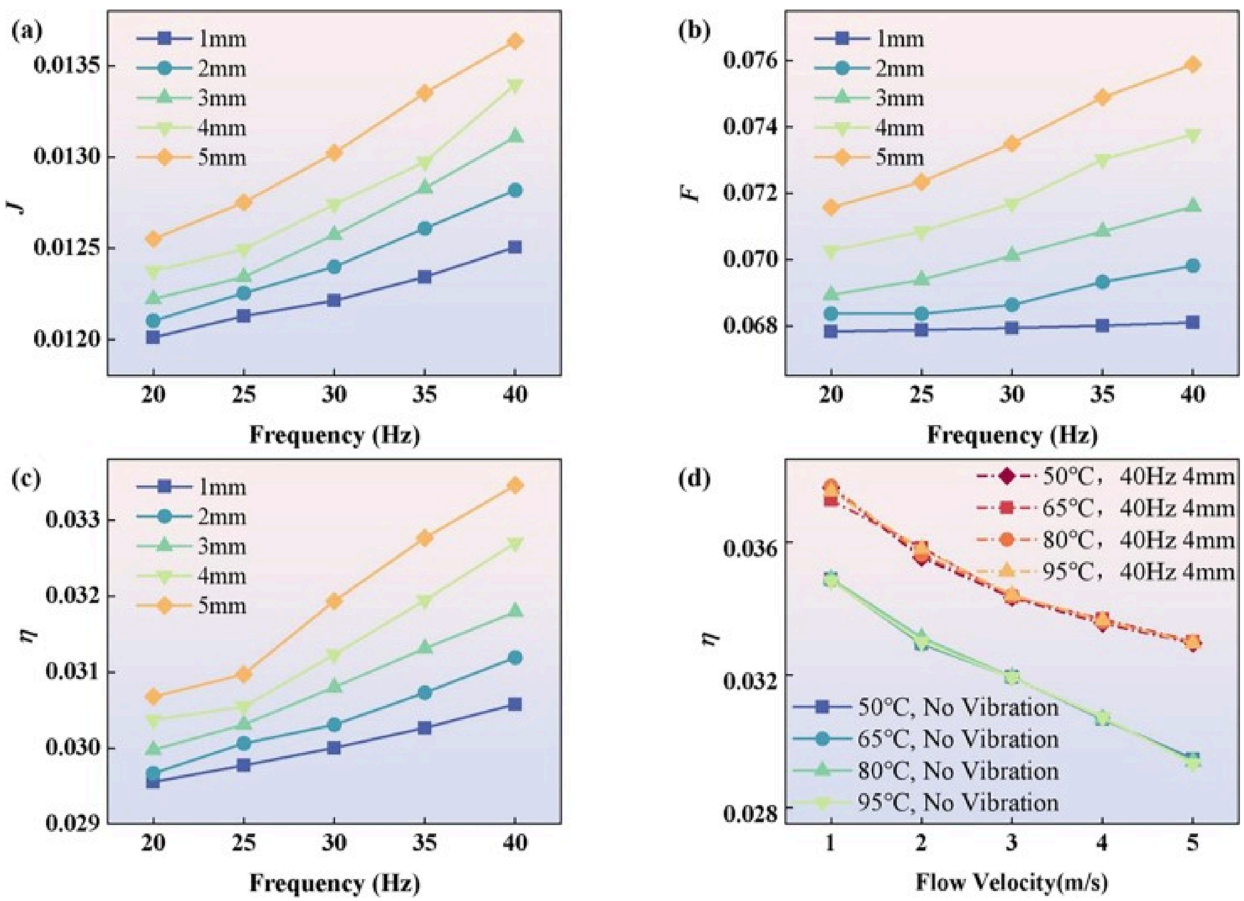

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Fig. 17. Dimensionless factor distribution of heat transfer in Kelvin metal foam under vibration: (a)Colburn J factor; (b) Flow friction factor F; (c) Integrated heat transfer factors η; (d) Comprehensive heat transfer factors at different heat source temperatures η.

Fig. 17(c) illustrates the distribution of η of the Kelvin metal foam heat exchanger under vibration conditions. A comparison with Fig. 11 reveals that the simulation results are consistent with the experimental results. The η shows a 4.3% and 13.2% increase under 40 Hz-1 mm and 40 Hz-5 mm conditions, respectively, compared to non-vibration conditions (η =0.0293). To reduce the effect of heat source temperature on its heat transfer performance, η for both vibration and non-vibration conditions were evaluated at different heat source temperatures, shown in Fig. 17 (d). Under different heat source temperatures, η decreases as wind speed increases. In non-vibration conditions, the maximum and minimum η at wind speed of 1 m/s to 5 m/s are 0.0348 and 0.0293, respectively. Under vibration conditions, η are 0.0375 and 0.0328. The average increase in comprehensive heat transfer factor η induced by vibration is 9.87%.

4. Conclusions

This work studies the effect of mechanical vibrations on the heat transfer performance of Kelvin open cell metal foam heat exchanger fabricated by selective laser melting via experimental and simulation methods. How vibrations improve the heat transfer performance of the metal foam heat exchanger is quantitatively analyzed, and the mechanisms by which variations in vibration amplitude and frequency contribute to improved heat transfer properties are elucidated.

Experimental results revel that mechanical vibrations can significantly increase the effective heat transfer coefficient (*EHTC*) of the Kelvin metal foam heat exchanger. Without vibration, the *EHTC* ranges from 333.69 W/(m^2·K) to 1366.1 W/(m^2·K) at air velocities between 1 m/s and 5 m/s. The *EHTC* increases to a maximum of 1556.37 W/(m^2·K) with an increase rate of 13.93% at vibration conditions within the frequency of 20–40 Hz and amplitudes of 1–5 mm. The Analysis of Variance reveals that amplitude and frequency similarly affect its heat transfer performance within the vibration range of 1–5 mm and 20–40 Hz. Experiments show that vibrations enhance heat transfer in Kelvin metal foam heat exchanger, increasing the Colburn *J* factor by up to

14.3% at 40Hz and 5mm. Moreover, the comprehensive heat transfer factor η improved by 9.87% in simulations and 11.2% in experiments, validating the simulation's accuracy.

A pore-scale vibration conjugate heat transfer model was developed to reveal how changes in vibration amplitude and frequency enhance heat transfer of Kelvin metal foam heat exchanger. Simulation results indicate that vibration periodically disrupts the flow field in the Kelvin metal foam heat exchanger, causing fluctuations in local velocity and the thermal boundary layer. The heat transfer coefficient shows a periodic response with a phase lag compared to the vibration velocity, changing at twice the vibration frequency. This work offers significant data reference for practical applications of Kelvin metal foam heat exchanger.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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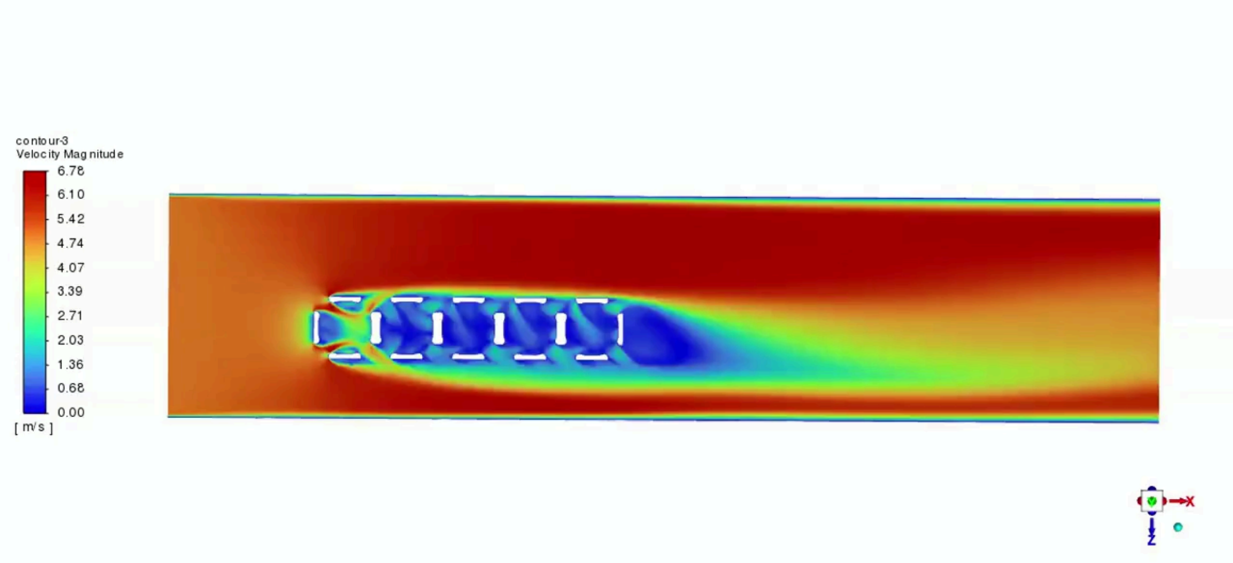
The following are the Supplementary data to this article:



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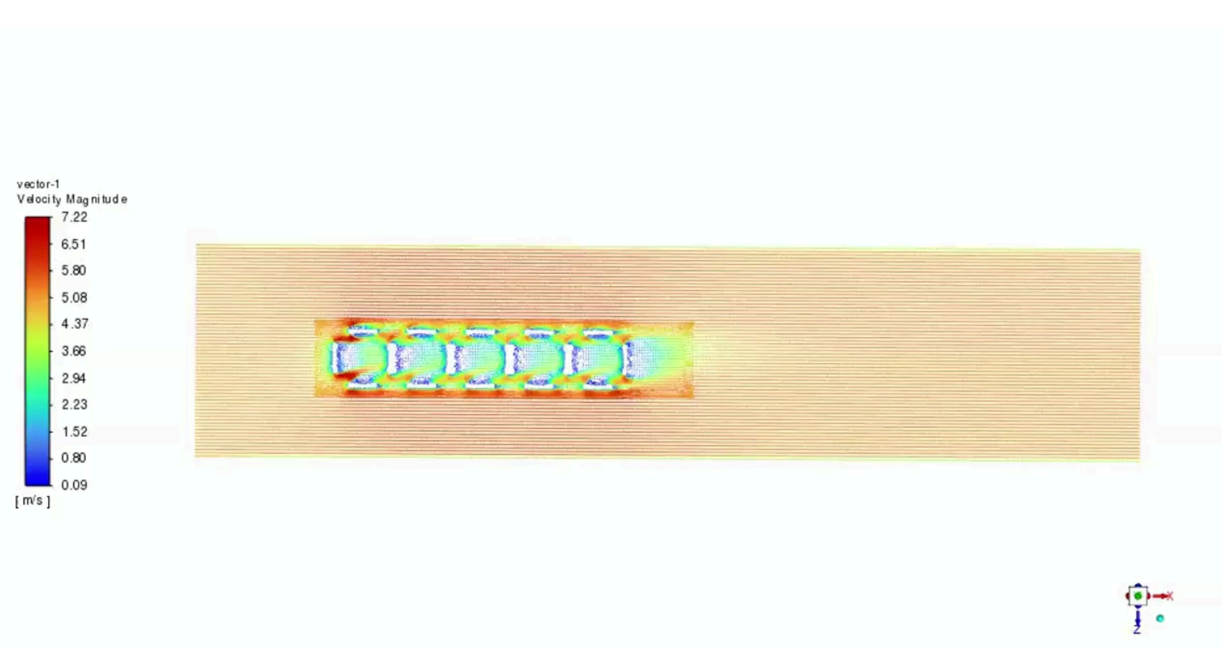

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Supplementary video 3.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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